

Introduction

Cluster analysis is a technique in which observations are partitioned into clusters.

There is a stigma around Taylor Swift’s music, which causes her music to be perceived as plain. In this project, cluster analysis is used to show that there exists variety within her style.

Dataset

The data used is a subset of Swift’s works, obtained from Kaggle (Priester, 2024).

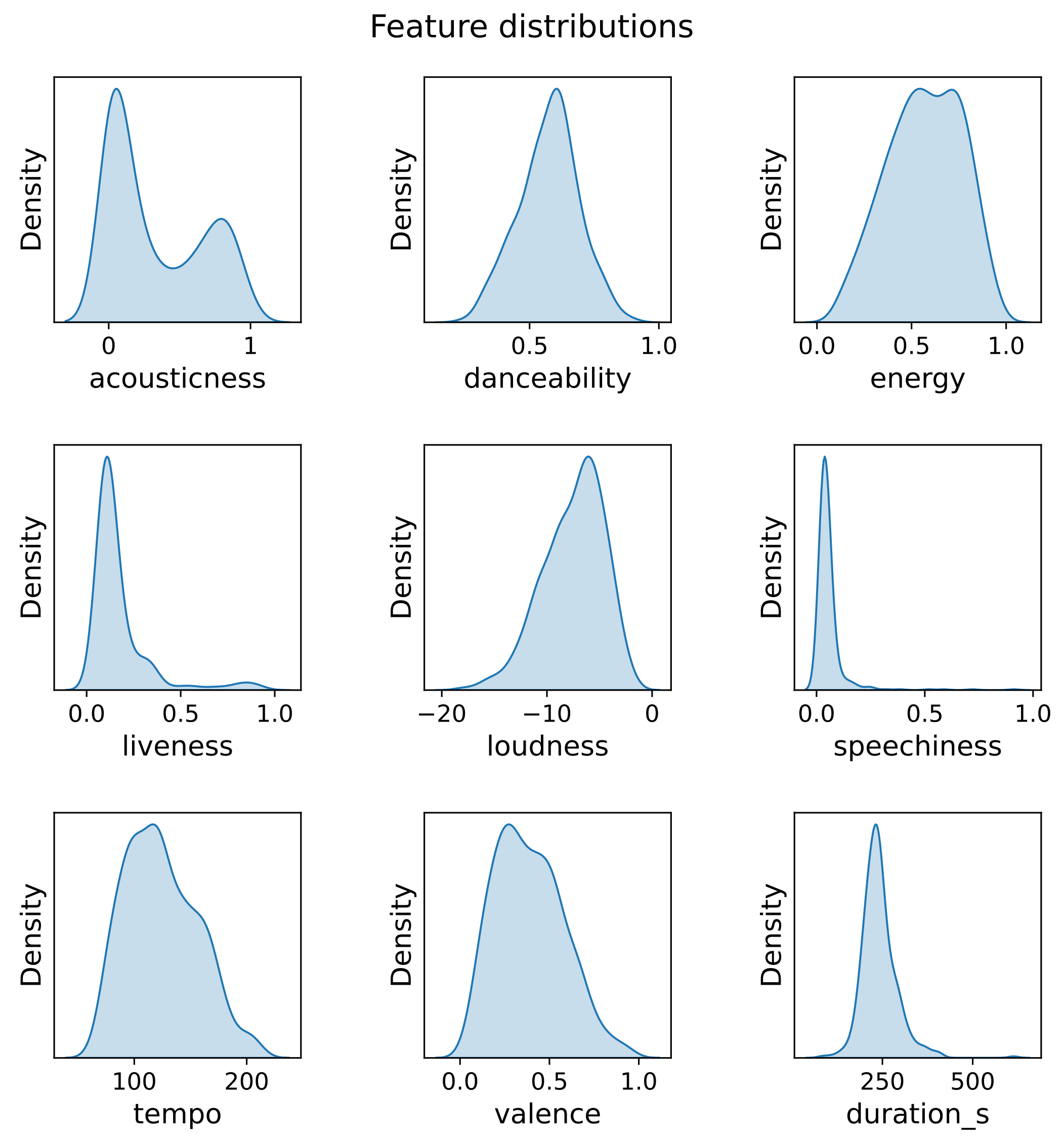


Figure 1: Density estimate plots of musical features.

Code

Clustering and visualisations were done in Python, using numpy, pandas, scikit-learn, matplotlib and seaborn. The code used can be found here: <https://github.com/lewistang-uk/Cluster-Analysis/>

Method

- Scale features to standard normal distribution.
- Select number of clusters using intrinsic metrics.
- Apply clustering algorithm using all features.
- Select features by analysing distributions across clusters.
- Apply clustering algorithm using selected features.
- Rename clusters by their main characteristic.

Selection of number of clusters

- Elbow (Thorndike, 1953)
 - Plot WCSS (Within-Cluster Sum of Squares) against number of clusters.
 - Find the point where the gradient drops sharply.
- Silhouette (Rousseeuw, 1987)
 - Measure the similarity of a point to its own cluster compared to other clusters.
 - Find the average of this metric and maximise by varying the number of clusters.

Clustering algorithms

- K-Means Clustering (MacQueen, 1967)
 - Initialise by selecting cluster centres (centroids).
 - Assign each observation to a cluster by its nearest centroid.
 - Adjust centroid position to minimise within-cluster variance.
 - Iterate until convergence.
- Agglomerative Clustering (Ward, 1963)
 - Initialise with each observation in its own cluster.
 - Recursively merge the most similar pair of clusters until the required number of clusters is obtained.

Application of clustering

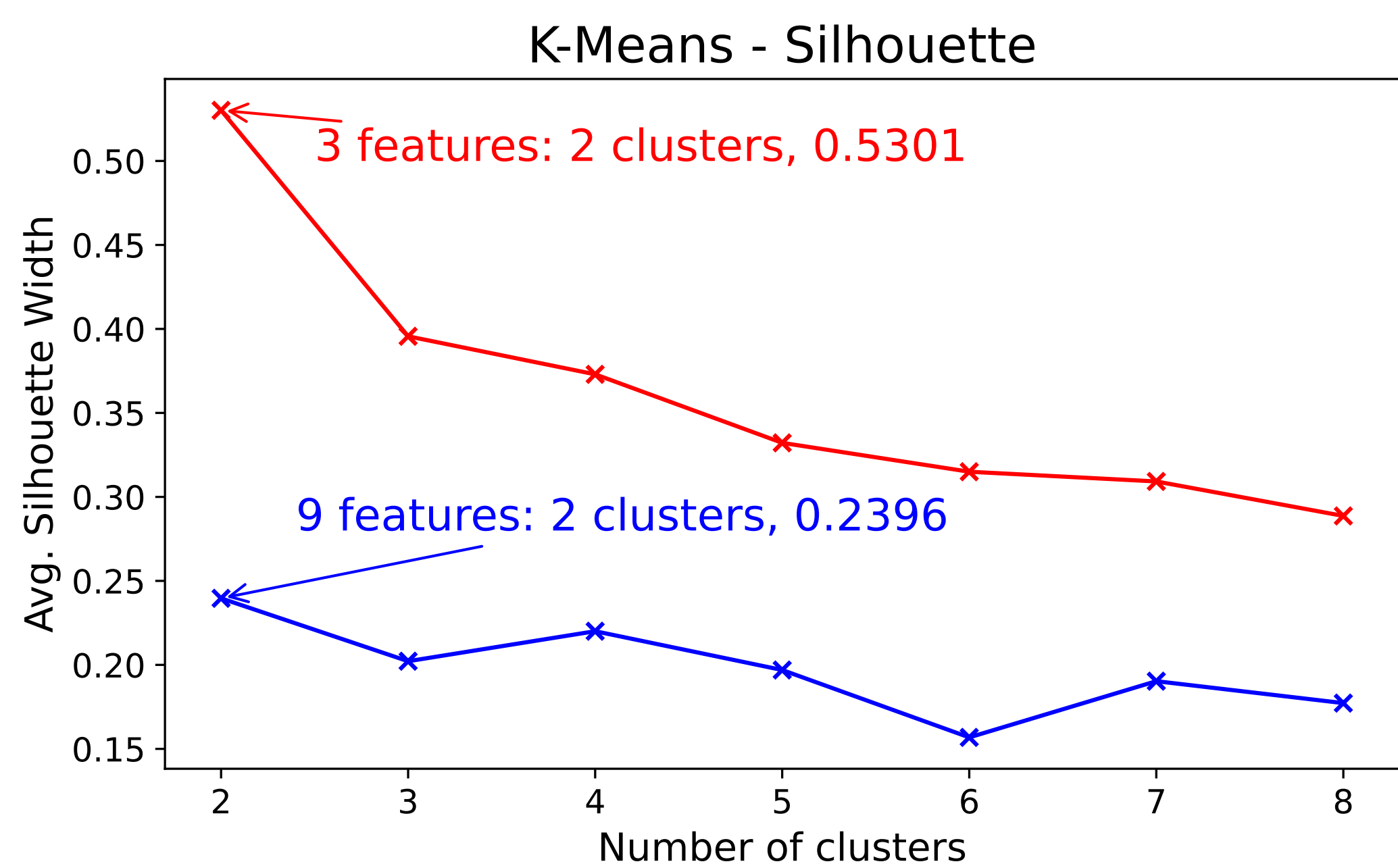


Figure 2: Feature selection reduces noise, creating clearer clusters. This significantly increases average silhouette width while suggesting the same number of clusters.

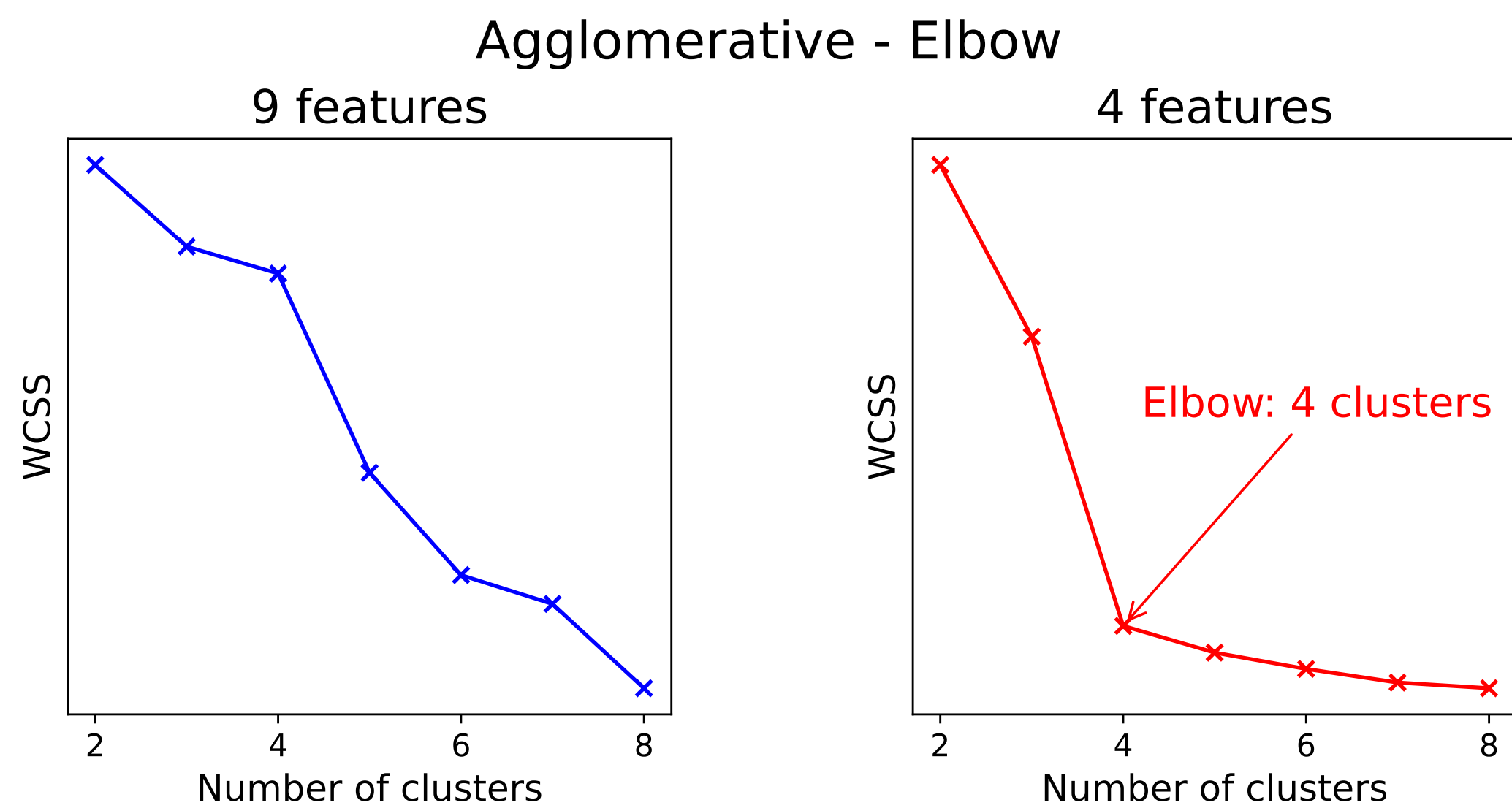


Figure 3: After feature selection, an elbow is present in the WCSS graph.

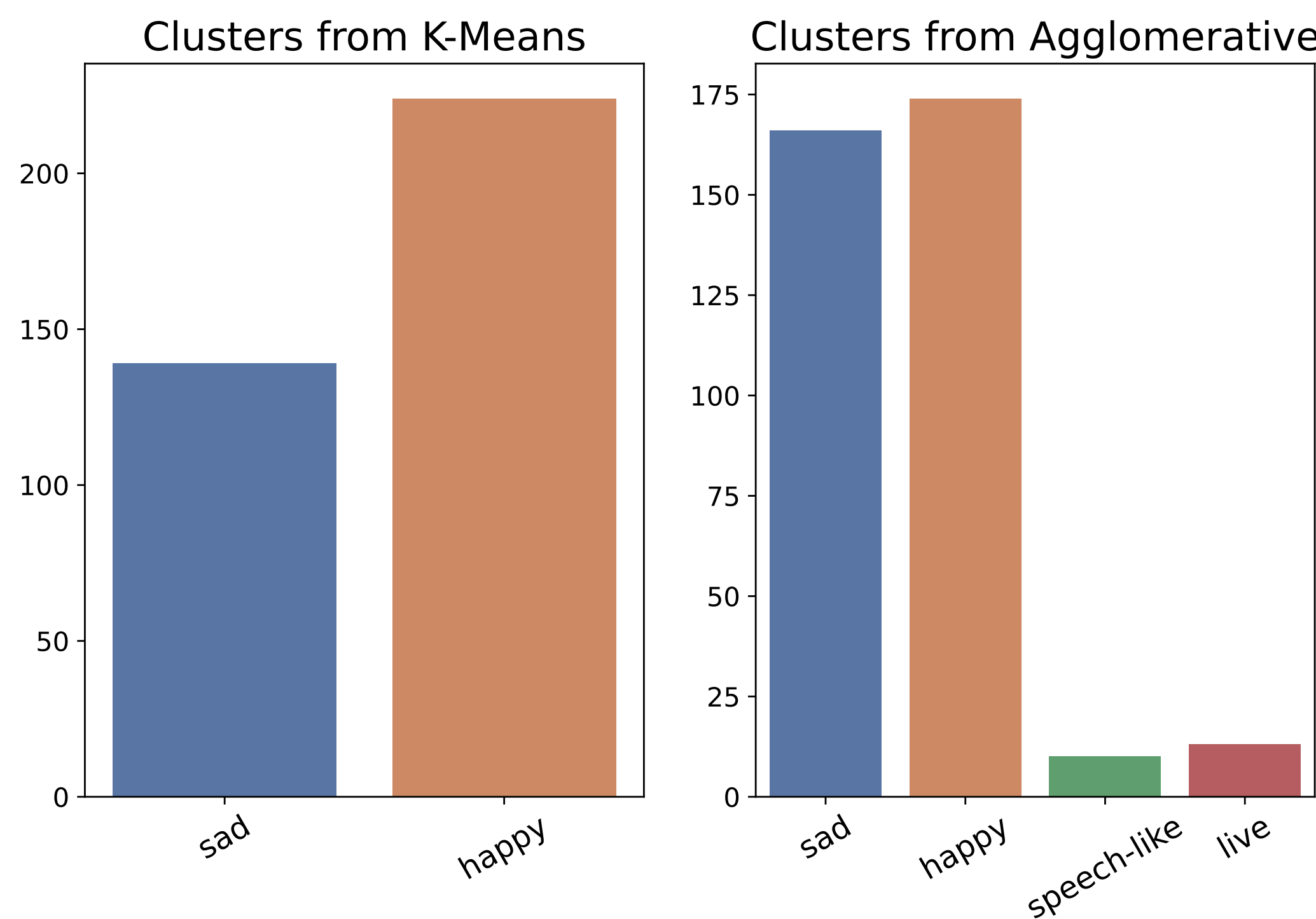


Figure 4: Number of observations in each cluster for both clustering algorithms.

Examples of differing cluster labels

song	K-Means	Agglomerative
closure	sad	speech-like
Treacherous	happy	sad
BTR - Live/2011	happy	live

Summary

- Agglomerative finds more hidden patterns – K-Means implicitly assumes that clusters are roughly equal in cardinality (Bishop, 2006).
- Silhouette is a better evaluation metric than Elbow – the implied optimum can always be found.
- Both clustering algorithms imply two main clusters, showing that there is some variety within Swift’s music.
- In an interview, Swift (2022) described three different styles of lyric writing. Analysis of lyrics could point to a different clustering.

References

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